

Constructing tomorrow with data



NFDI4ING

Research data management
in engineering sciences

NFDI4ING's RDM Basics for Engineers

Teil 2 von 4: **Best Practices in Research Data Management: DMPs, RDMO und Best Practice im Umgang mit Forschungsdaten**

30.03.2026, online

NFDI4ING (National Research Data Infrastructure for Engineering Sciences)

Isabel Hohle, Michaela Leštáková
ULB Darmstadt/ TU Darmstadt

Agenda

RDMO

NFDI4ING

RDMO NFDI4ING

Forschungsdatenmanagement gemeinsam an einem Ort planen

Teil I

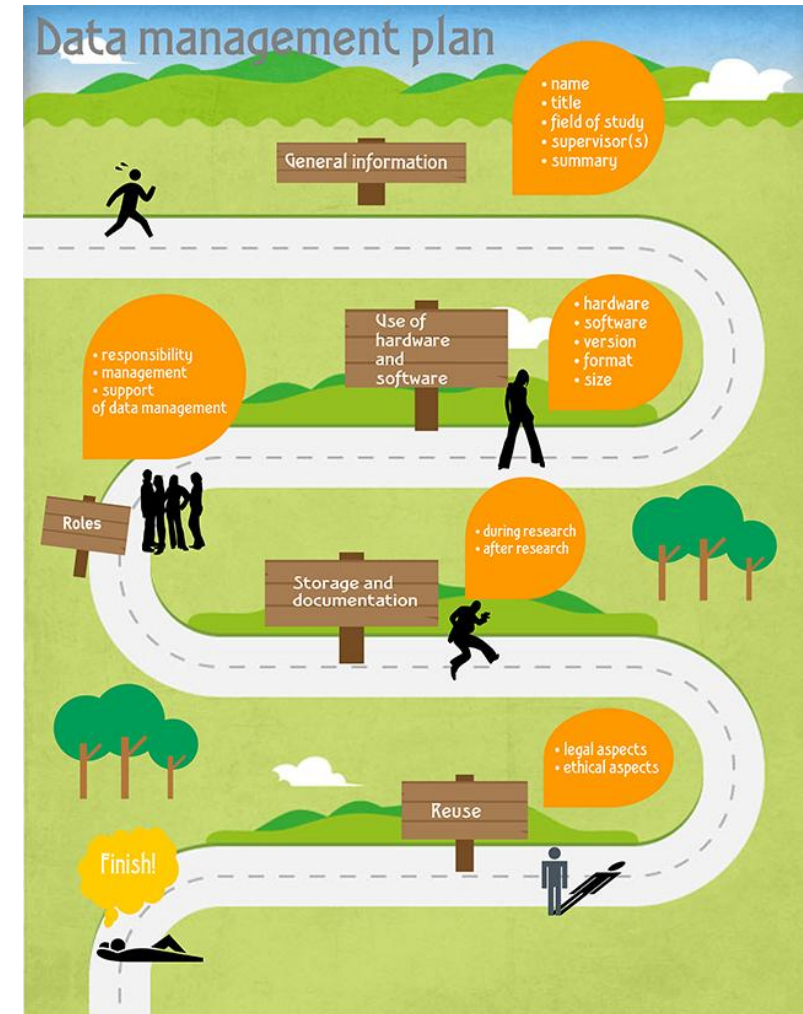
- Forschungsdatenmanagement und DMPs
- Einführung in RDMO und RDMO NFDI4ING
- Kurze Live Demo RDMO

Teil II

- Best Practice im Umgang mit Forschungsdaten
- Übung: Erstellung eines eigenen DMPs in RDMO

Teil III

- Fragen & Feedback



<https://libguides.ru.nl/datamanagementEN>

Teil I

Forschungsdatenmanagement und DMPs

RDMO

Herausforderungen des FDM



Diverse Data: Die ingenieurwissenschaftliche Forschung generiert heterogene Daten (Simulationen, Experimente, Software usw.).



Compliance: Förderorganisationen (DFG, EU usw.) verlangen zunehmend Datenmanagementpläne.



FAIR Prinzipien: Daten sollten auffindbar, zugänglich, interoperabel und wiederverwendbar sein.



Zusammenarbeit: Teams benötigen gemeinsame, versionsverwaltete und zugängliche Datenmanagement-Lösungen, Interdisziplinarität, die Einbindung der Industrie und Kooperationen.



Zeitdruck: Forschenden fehlt es oft an Zeit und Fachwissen für eine umfassende RDM-Planung.

Was ist ein Datenmanagementplan (DMP)?

- Ein DMP beantwortet zentrale Fragen zu Forschungsdaten, z. B.



projektbezogene
administrative
Informationen



Rollen,
Verantwortlichkeiten
und Verpflichtungen



Budget, Kosten,
Ressourcen



Datentypen und -
formate



Metadaten (-Standards),
Nachweisbarkeit



Zugang zu Daten
und deren
Publikation



Datenaufbewahrung



Datenlöschung

DMPs – Anforderungen von Förderinstitutionen

Wird ein DMP gefordert?



(abhängig vom Förderprogramm)



Umgang mit Forschungsdaten

Checkliste für Antragstellende zur Planung und zur Beschreibung des Umgangs mit Forschungsdaten in Forschungsvorhaben

Diese Checkliste unterstützt Sie, wesentliche Aspekte des Umgangs mit Forschungsdaten strukturiert zu beschreiben sowie die für die Umsetzung benötigten Ressourcen und Kompetenzen sichtbar zu machen. Bitte nehmen Sie zu den unten genannten Themenfeldern im Antrag unter Punkt 2.4 Stellung.

DFG Checkliste

EU Grants: Data management plan (HE):v1.1 – 01.04.2022

DATA MANAGEMENT PLAN

(To be filled in and uploaded as deliverable in the Portal Grant Management System, at the due date foreseen in the system (and regularly updated).)

⚠ The template is recommended but not mandatory. If you do not use it, please make however sure that you comply with the research data management requirements under Article 17 of the Grant Agreement.)

PROJECT	
Project number:	[project number]
Project acronym:	[acronym]
Project name:	[project title]

DATA MANAGEMENT PLAN	
Date:	[dd/mm/yyyy]
Version:	[DMP version]

1. Data Summary

Will you re-use any existing data and what will you re-use it for? State the reasons if re-use of any existing data has been considered but discarded.

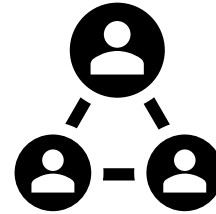
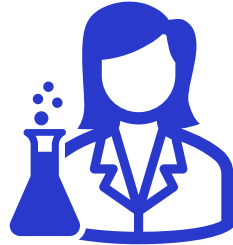
What types and formats of data will the project generate or re-use?

What is the purpose of the data generation or re-use and its relation to the objectives of the project?

What is the expected size of the data that you intend to generate or re-use?

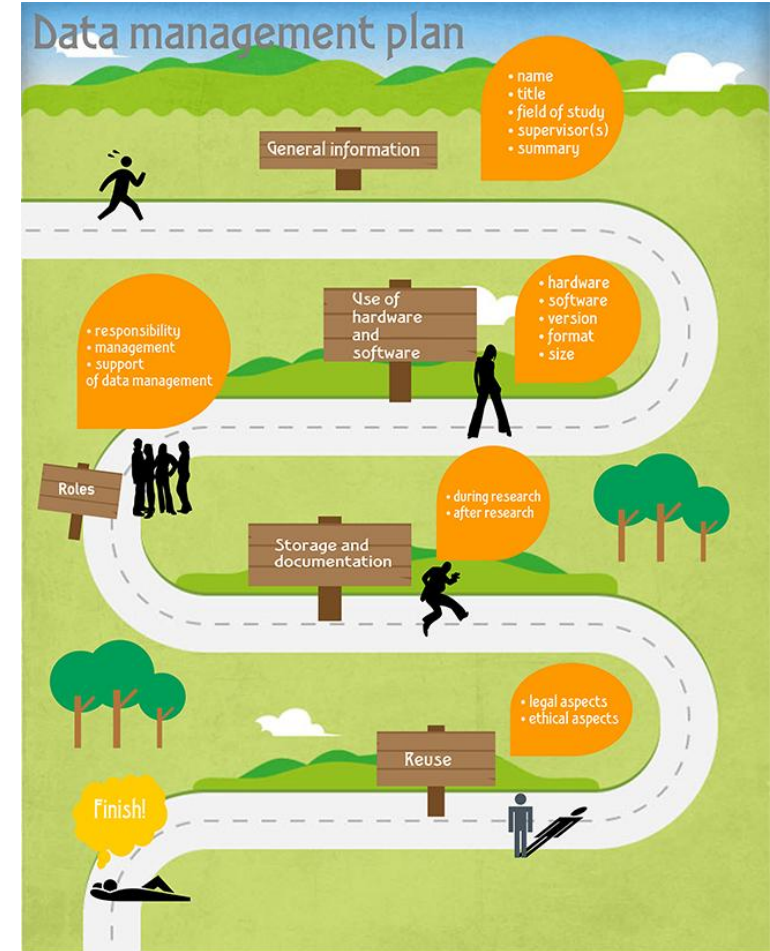
DMP Horizon Europe

DMPs – Roadmap und Planungstool



Checkliste, Roadmap und Unterstützung bei der Planung des eigenen Projekts von Anfang bis Ende

Nützliche Übersicht zum Beispiel bei forschungsdaten.info



<https://libguides.ru.nl/datamanagementEN>

Datenmanagementpläne



Vorteile

Strukturierte Planung: Klare Roadmap für das Datenmanagement im Projekt

Förderkonformität: Erfüllt Anforderungen von Fördergebern und Institutionen

Effiziente Dokumentation: Vereinfacht Berichte und Nachweise

Nachnutzbarkeit: Eigene Daten leichter wiederverwendbar durch klare Strukturen

Mehr Sichtbarkeit: Höhere Zitationschancen durch geregelte Nachnutzung

Datenpublikation: Gilt als eigenständige wissenschaftliche Leistung

Datensicherheit: Reduziert Risiko von Datenverlust

Langfristige Lesbarkeit: Erhöht Chance auf lesbare Formate nach Jahren

Wissenssicherung: Erleichtert Wissenstransfer bei Personalwechsel



Nachteile

Ressourcenaufwand: Überwachung und Kommunikation erfordern Personal

Zeit- und Kostenfaktor: Erstellung und Umsetzung brauchen Zeit und ggf. Infrastruktur

(Technische Anpassungen: Eventuell Konvertierung bestehender Dateiformate nötig)

(forschungsdaten.info)

Unterstützung durch ein DMP tool

- Unterstützung bei der Planung und Dokumentation von FDM-Prozessen
- Angepasst an spezifische Fachbereiche und Anforderungen
- Integration in bestehende Arbeitsabläufe (Repositorien, Labordatenbanken)
- Benutzerfreundlich und einfach zu bedienen
- Geeignet sowohl für FDM-Einsteiger als auch für Experten



DMP4NFDI – NFDI Basic Service

Support the development and provision of standardized Data and Software Management Plan
(DMP/SMP) services within the NFDI.



Hosting
&
Integration with other
services



Template standardisation
&
Support for development



Training
&
Support for outreach

Datenmanagementpläne und RDMO



- Wer hat schon einmal einen Datenmanagementplan erstellt?



- Wer hat schon Erfahrungen mit RDMO?

RDMO – Research Data Management Organiser

Was ist RDMO ?



[Feedback](#)

[Language](#) ▼

RDMO NFDI4ING

Planning research data management collaboratively in one place

Discover Supercomputer 3 (NASA Goddard Space Flight Center) / CC BY 2.0

Welcome to RDMO

The tool enables engineering researchers to create a data management plan (DMP) for their research projects.

RDMO guides you through all aspects of research data management using questionnaires and offers various export formats for downloading the collected information, e.g. for use in your funding application process.

[NFDI AAI with Unity](#)

Questions or feedback?

[Contact us](#) for any questions, feedback and support.

Maintenance window

Tuesday 6:30 am - 8:30 am

RDMO – Research Data Management Organiser

- **Open-source web-based tool** zur Erstellung und Verwaltung von DMPs sowie zur Unterstützung von Forschenden bei allen Aspekten des Forschungsdatenmanagements
- **RDMO NFDI4ING:** Der für die Ingenieurwissenschaften angepasste Client von NFDI4ING mit speziellen Katalogen und Anleitungen



Community und Verein

<https://rdmorganiser.github.io/>

NFDI4ING Feedback Language Isabel Hohle

My research project

Description	No description available.	
Catalog	NFDI4ING DMP Template (Version 2) Catalog according to the DFG checklist supplemented by engineering content.	

Options

Answer questions

View answers

- Update project information
- Update project catalog
- Update parent project
- Copy project
- Delete project

Add member

Create snapshot

Back to projects overview

Export

RDMO XML

CSV (comma separated)

CSV (semicolon separated)

JSON

as ROCrate JSON

Import values

Import from file

Select file

Tasks

Tasks are generated automatically from the answers given in the project. On the page of each task you can see which of your answers lead to the activation of the task.

No active tasks found.

Members

Here you can see who can access the project and invite additional members. You can use the user roles to manage which rights the users have. Unless you are the last owner, you can leave the project with the button next to your name.

User	E-Mail	Role	
Isabel Hohle	isabel.hohle@tu-darmstadt.de	Owner	

Snapshots

Snapshots allow you to save all responses at a given point in time and preserve a certain stage of the project. Later the snapshot can be used to create views, and the project can also be reset to a previous snapshot if needed.

Snapshot	Description	Created	
First Project Snapshot		March 5, 2026, 3:03 p.m.	

RDMO Service

Zahlreiche Institutionen bieten eigene RDMO Instanzen an

Aber auch multi-site Hosting mit RDMO clients für verschiedene Mandanten

RDMO
Sprache ▾ Anmeldung

Willkommen bei RDMO

der zentralen Plattform der Max Weber Stiftung für das Datenmanagement

Mit RDMO (Research Data Management Organizer) steht Ihnen ein Werkzeug zur Verfügung, welches Sie bei der strukturierten Erstellung von Datenmanagementplänen und Textbausteinen für Ihren Förderantrag unterstützen wird. Die verschiedenen Schritte des Fragebogens werden dabei Schritt für Schritt durch den RDMO geleitet.



ZIB
ZUSE INSTITUTE BERLIN

Research Data Management Organizer

Anmeldung

Benutzername:

 Passwort:

Englisch Deutsch

Anmeldung

Willkommen bei RDMO

Der Research Data Management Organizer unterstützt Euch bei der Erstellung strukturierter Datenmanagementpläne. Idealerweise habt Ihr ja jetzt schon ein paar Wochen in Eurem neuen Projekt zugebracht und ein Gefühl dafür, was an Datenhandling auf Euch zukommt.

Wenn Euer Projekt Teil eines übergeordneten Projekts ist, besteht die Möglichkeit den DMP des übergeordneten Projekts zu übernehmen und ggf. anzupassen. Meldet Euch in dem Fall gerne unter der unten angegebenen Email.

Ansonsten startet Ihr mit einem leeren Formular, was aber auch harmlos ist und zum Bearbeiten nicht länger als eine halbe Stunde dauern sollte.

Willkommen bei RDMO an der TU Braunschweig

Diese Webseite ist eine Testinstallation zur Erstellung und Pflege von Datenmanagementplänen für Mitglieder der TU Braunschweig und deren Kooperationspartner.

Anmelden

Mit Shibboleth anmelden

[rdmo.h-da.de](#)
[Language ▾](#)
[Login](#)

RDMO

The tool to support the planning and implementation of research data management at Hochschule Darmstadt.

[TU dmo](#)
[Feedback](#)
[Sprache ▾](#)

TUdmo

Forschungsdatenmanagement gemeinsam an einem Ort planen

[Account](#)

[TU Clausthal](#)
[Feedback](#)
[Sprache ▾](#)

RDMO TU Clausthal

Das Werkzeug zur Unterstützung von Planung und Umsetzung des Forschungsdatenmanagements an der TU Clausthal

Willkommen bei TUdmo

Der TU data management organizer (TUdmo) unterstützt Forschende der TU Darmstadt bei der Planung und Umsetzung von Forschungsdatenmanagementplänen (DMPs) für Forschungsprojekte.

Anhand eines Fragebogens leitet TUdmo Sie durch die verschiedenen Aspekte des Forschungsdatenmanagements. Sie können Ihre Daten nach Ihren Anforderungen anschließend verschiedene Exportmöglichkeiten, um die erhobenen Informationen zu veröffentlichen.

TUdmo basiert auf der Open Source Software RDMO, die an der ULB Darmstadt mit der TU Clausthal entwickelt wurde.

Willkommen bei RDMO

Als Mitglied der TU Clausthal können Sie das Werkzeug RDMO nutzen, um die Planung und Umsetzung von Forschungsdatenmanagementplänen (DMPs) zu planen und zu realisieren. Sie können Ihre Daten nach Ihren Anforderungen anschließend verschiedene Exportmöglichkeiten, um die erhobenen Informationen zu veröffentlichen.

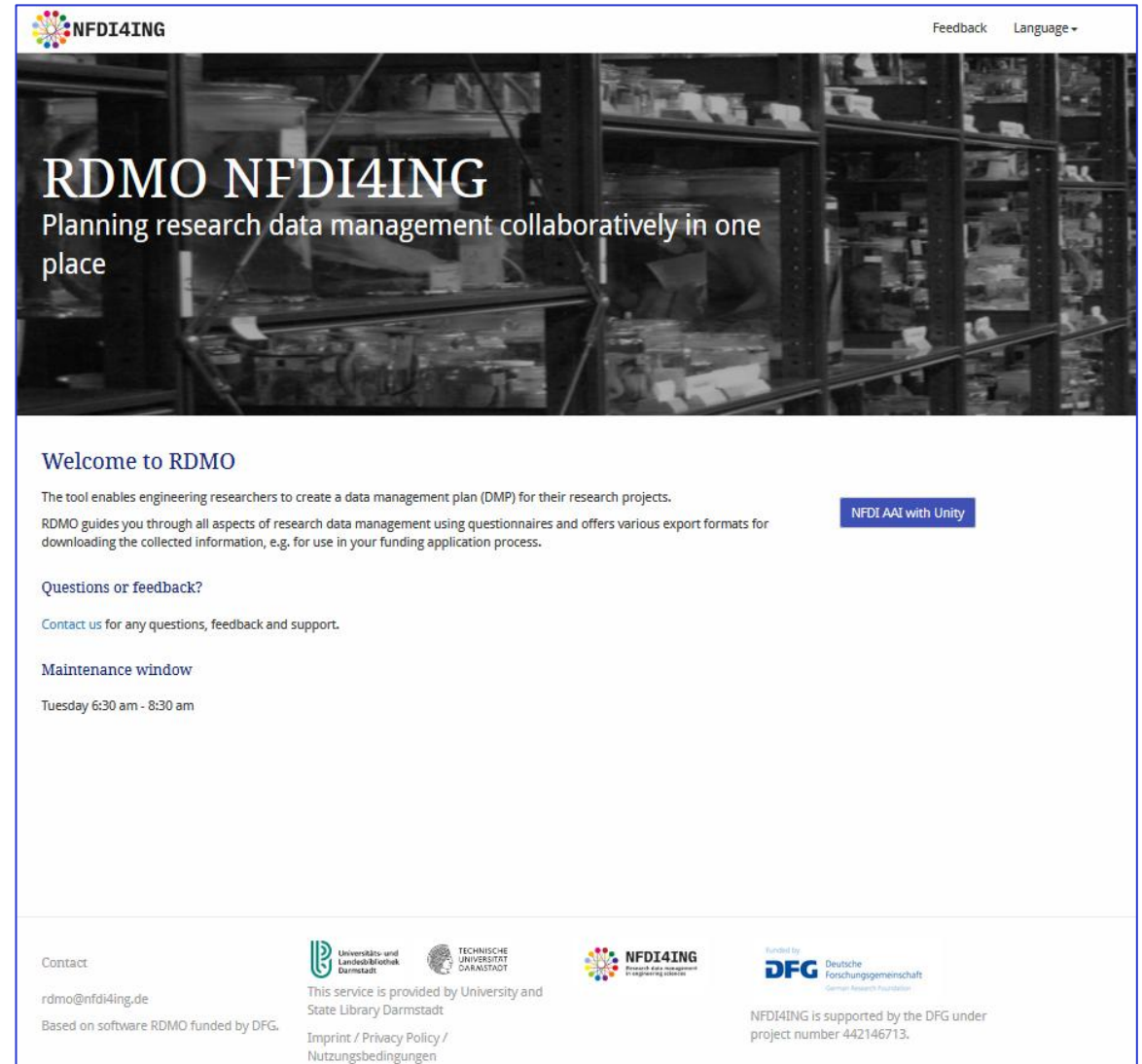
NFDI4ING RDMO Service

rdmo.nfdi4ing.de

- Login via NFDI-AAI

Kataloge:

- NFDI4ING DMP Template
- Software Management Plan



The screenshot shows the homepage of the NFDI4ING RDMO Service. At the top, there is a header with the NFDI4ING logo on the left and links for 'Feedback' and 'Language' on the right. Below the header is a large banner image of a library shelf filled with books. Overlaid on the banner is the text 'RDMO NFDI4ING' in large white letters, followed by the subtitle 'Planning research data management collaboratively in one place' in smaller white text. Below the banner, the main content area has a heading 'Welcome to RDMO' in blue. Underneath, a paragraph explains that the tool enables engineering researchers to create a data management plan (DMP) for their research projects. To the right of this paragraph is a blue button labeled 'NFDI AAI with Unity'. Below the paragraph, there is a link 'Contact us' for questions, feedback, and support. Further down, a section titled 'Maintenance window' indicates a Tuesday 6:30 am - 8:30 am downtime. The footer contains contact information (rdmo@nfdi4ing.de), logos for the University and State Library of Darmstadt, Technische Universität Darmstadt, and NFDI4ING, and a DFG logo with text about funding by the Deutsche Forschungsgemeinschaft under project number 442146713. It also includes links for 'Imprint / Privacy Policy / Nutzungsbedingungen'.

NFDI4ING

Feedback Language

RDMO NFDI4ING

Planning research data management collaboratively in one place

Welcome to RDMO

The tool enables engineering researchers to create a data management plan (DMP) for their research projects. RDMO guides you through all aspects of research data management using questionnaires and offers various export formats for downloading the collected information, e.g. for use in your funding application process.

[NFDI AAI with Unity](#)

[Questions or feedback?](#)

[Contact us](#) for any questions, feedback and support.

Maintenance window

Tuesday 6:30 am - 8:30 am

Contact

rdmo@nfdi4ing.de

Based on software RDMO funded by DFG.

 Universitäts- und Landesbibliothek Darmstadt

 TECHNISCHE UNIVERSITÄT DARMSTADT

This service is provided by University and State Library Darmstadt

 **NFDI4ING**
Research data management
in engineering sciences

 **DFG** Deutsche Forschungsgemeinschaft
German Research Foundation

NFDI4ING is supported by the DFG under project number 442146713.

Imprint / Privacy Policy / Nutzungsbedingungen

RDMO Workflow

- Login via NFDI-AAI
- Create a new project
- Invite members
- Start interview
- Add datasets
- Answer questions
- Share or export DMP

 **NFDI4ING**

Feedback Language▼ Isabel Hohle▼

My projects

Import project + New project

1 of 1 projects are displayed

✕ Search

Show filters

Name ⇅	Progress ⇅	Role ⇅	Last changed ⇅	
My research project NFDI4ING DMP Template (Version 2)	19 of 42	Owner	Jan 29, 2026, 10:35 AM	  

RDMO Workflow

- **Create a new project**

- Title and description
- Choose a catalog = Interview guide
- Different templates available for different use cases
- Link to a parent project if needed

 Feedback

Create new project

Title
The title for this project.

Description
A description for this project (optional).

Catalog
The catalog which will be used for this project.

☒ **NFDI4ING DMP Template (Version 2)**
Catalog according to the DFG checklist supplemented by engineering content.

☐ **Software Management Plan**
This catalogue is for the management of scientific software projects. It supports scientists in the development and project organisation of software developments through fifty questions in different topic blocks. This is version 3.0 of the catalogue.

☐ **Project specific: SPP2419**
This catalog was created for the DFG SPP2419 HyCAM.

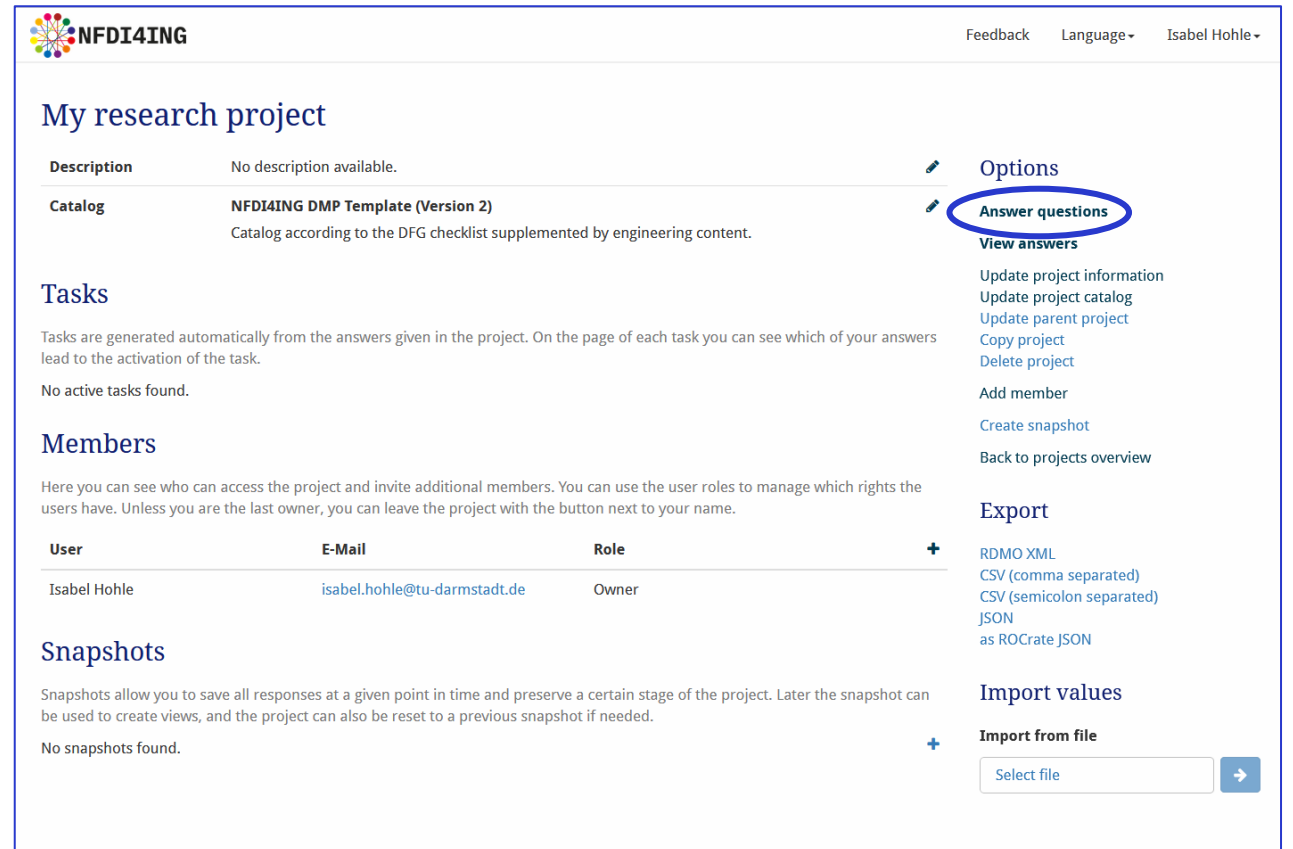
Parent project
The parent project of this project.

Create project Cancel

RDMO Workflow

Project overview

- Invite members to the project
- Create versions (snapshots)
- Change settings (e.g. catalog)
- Start the interview with „Answer questions“



NFDI4ING Feedback Language Isabel Hohle

My research project

Description	No description available.
Catalog	NFDI4ING DMP Template (Version 2) Catalog according to the DFG checklist supplemented by engineering content.

Options

- Answer questions** (circled in blue)
- View answers
- Update project information
- Update project catalog
- Update parent project
- Copy project
- Delete project
- Add member
- Create snapshot
- Back to projects overview

Export

- RDMO XML
- CSV (comma separated)
- CSV (semicolon separated)
- JSON
- as ROCrate JSON

Import values

Import from file

Select file

Tasks

Tasks are generated automatically from the answers given in the project. On the page of each task you can see which of your answers lead to the activation of the task.

No active tasks found.

Members

Here you can see who can access the project and invite additional members. You can use the user roles to manage which rights the users have. Unless you are the last owner, you can leave the project with the button next to your name.

User	E-Mail	Role	
Isabel Hohle	isabel.hohle@tu-darmstadt.de	Owner	+

Snapshots

Snapshots allow you to save all responses at a given point in time and preserve a certain stage of the project. Later the snapshot can be used to create views, and the project can also be reset to a previous snapshot if needed.

No snapshots found.

NFDI4ING DMP Template

- For researchers of engineering sciences
- based on RDA survey and community workshops
- follows DFG checklist:
https://www.dfg.de/research_data/checklist

The screenshot shows the 'Data Quality' section of the NFDI4ING DMP Template. The page has a header with the NFDI4ING logo, 'Back to project', and user information 'Isabel Hohle'. A breadcrumb trail reads 'My Projects / My research project / Documentation and data quality'. The main heading is 'Data Quality', followed by instructions: 'Please fill in the form for each tab. The same tabs may be used later on other pages. You can add a new tab using the green button. Once created, you can edit or delete tabs using the buttons in the top right corner.' Below this is a tab labeled '1 Dataset' with a '+ Set' button. The main content area is titled 'What measures are taken to ensure high quality of the data?' and includes a sub-header 'The quality of the data can be secured by various measures and depends on your engineering sub-discipline.' A list of checkboxes follows: 'Using a calibration method' (checked), 'Use of calibrated measuring equipment' (checked), 'Repetitions of measurements' (checked), 'Validation of data input and data output' (unchecked), 'Validation of the evaluated data using external models/benchmarks or scientific experiments' (unchecked), 'Use of quality metrics for images' (unchecked), 'Specification of the model uncertainty' (unchecked), 'Documentation of the relevant parameters and metadata' (unchecked), 'Calibration of measuring devices' (checked), 'Carrying out measurements according to the standard' (unchecked), and 'Use of predefined quality criteria. These include:' (unchecked). Below the last checkbox is a large text input field. At the bottom, there is an 'Other:' label and a text input field. On the right side, there is a sidebar with 'Overview' (Project: My research project, Catalog: NFDI4ING DMP Template (Version 2), Reload page, Back to my projects), 'Progress' (19 of 42, Back, Proceed), and 'Navigation' (Introduction, Data description (9 of 12), Documentation and data quality (4 of 8), Documentation (2 of 4), → Data Quality (2 of 4), Storage and technical backup during th..., Data exchange and permanent accessib..., Legal obligations and framework condit..., Responsibilities and resources).

The Interview

Topics

- Data description
- Documentation
- Quality measures
- Storage and backup
- Data exchange
- Legal obligations, responsibilities and resources

NFDI4ING Back to project Feedback Language Isabel Hohle

My Projects / My research project / Introduction

Please add a dataset to your plan

The present catalog is the [checklist published by the DFG](#) supplemented by engineering content. The questions from the DFG checklist must be taken into account when submitting an application and completing the project and should help you to structure your data management.

During the interview, you will be asked several questions about the various aspects of data management. Depending on your research project, you may work with different types of data, so it might be necessary to plan different measures for each. Therefore, create as many sets as it makes sense for your approach.

A logical unit for your research data could be, for example, based on

- the methodology (e.g. simulation)
- data type (e.g. image, text, audio, video)
- accessibility of the data (e.g. open access, restricted access)

You must create at least one set. Please fill in the form for each set. Once created, you can edit it.

Create tab

Name

Simulation

Please give the tab a meaningful name.

Reuse answers

Search for project or snapshot title, or answer text ...

Restrict the search to a particular project ...

☐ Include snapshots in the search

You can populate this tab with answers from a similar tab in any project you are allowed to access.

Close Create

Overview

Project: [My research project](#)
Catalog: NFDI4ING DMP Template (Version 2)
[Show management panels](#)
[Reload page](#)
[Back to my projects](#)

Progress

0 of 4

Back Proceed

Navigation

Grey entries will be conditionally skipped based on your input.

Ask Chatbot

Introduction

The Interview

Export your DMP

- Available in different formats
- Full export or „views“ for specific templates

 **NFDI4ING** [Back to project](#) [Feedback](#) [Language](#) [Isabel](#)

Answers for *My research project*

In the following, we have summarized the information about the project as given by you and your collaborators.

Introduction

Please add a dataset to your plan

Data description

Content data description

What kind of dataset is it?

Set "Dataset":

- tables
- Measurement data e.g. from test benches
- image data
- visual recordings / video data

How is new data created in your project?

Set "Dataset": programming

Will existing data be reused?

Set "Dataset":

- Measurement data e.g. from test benches
- Reuse of data provided by industry partner

Technical data description

Which data types, in terms of data formats, are created in your project?

Set "I":

- (Drawing Exchange Format) *.dxf
- (TansForm Formular) *.tiff, *.tff

Set "Dataset":

- (Drawing Exchange Format) *.dxf
- (TansForm Formular) *.tiff, *.tff

How is the data processed in your project?

To what extent are these incurred for your data set or what data volume can be expected for the data set?

Set "Dataset": not yet defined

Documentation and data quality

Options

[Back to project overview](#)

Export

- PDF
- Rich Text Format
- Open Office
- Microsoft Office
- HTML
- Markdown
- mediawiki
- LaTeX

Software Management Plan (SMP)

NFDI4ING

Create new project

Title
The title for this project.

Description
A description for this project (optional).

Catalog
The catalog which will be used for this project.

☐ NFDI4ING DMP Template (Version 2)
Catalog according to the DFG checklist supplemented by engineering content.

☒ **Software Management Plan**
This catalogue is for the management of scientific software projects. It supports scientists in the development and project organisation of software developments through fifty questions in different topic blocks. This is version 3.0 of the catalogue.

☐ Project specific: SPP2419
This catalog was created for the DFG SPP2419 HyCAM.

Parent project
The parent project of this project.

Code

Which programming language(s) do you plan to use?

The software languages go hand in hand with different methods, focuses and required skill levels. Here you enter the language(s) you choose and why. Examples of software languages can be Java, Python, R, etc.

At this point it is also useful to document whether technical standards are relevant for the development. Examples: IEC Standard 62304 for software in medical applications, or ECSS-E-ST-40C for software in space applications, or exchange of medical image data.

Please enter your entries line by line. You can add lines using the green button and remove them using the blue cross (x).

Which technology or process is used for versioning?

Versioning code during development is strongly recommended. Applications such as Git systems are widespread.

My Projects / My Research / General

Topic

If you are looking for an introduction to the topic before completing a software management plan, then we recommend Martinez-Ortiz et al. (2022): Practical guide to Software Management Plans, v1.0, <https://doi.org/10.5281/zenodo.7589725>.

What is the title of the software project?

The title of the software can, of course, still change or be specified in the course of a project. Nevertheless, it makes sense to give the project a specific name at the beginning in order to facilitate further (internal) communication about it.

Which research field(s) does this software belong to?

The list of disciplines follows the subject classification of the DFG (German Research Foundation). Please enter your entries line by line. You can add lines using the green button and remove them using the blue cross (x).

In which application class is the software categorised?

The application classes are essentially based on the "DLR Software Engineering Guidelines" (<https://doi.org/10.5281/zenodo.1344612>), p. 7-8. Depending on the selection in this questions a selection of different questions is provided below.

Click here for more information on the application classes for research software

☐ Application class 0: The focus of the software is on personal use in conjunction with a small scope. The distribution of the software within and outside the own institution is not planned.

☒ Application Class 1: The software is only developed within a narrow scope. It is to be further developed and used beyond personal purposes.

☐ Application Class 2: The software is intended to ensure long-term development and maintainability. It is the basis for a transition to product status.

☐ Application Class 3: For the software it is essential to avoid errors and to reduce risks. This applies in particular to critical software and that with product characteristics.

What is the intended use of the software? How will your software contribute to research?

The intention for developing software in a scientific context can be manifold. The same applies to the specific application of the software and the contribution to scientific knowledge. Depending on the scientific community, these points can also differ significantly from each other.

We offer here some recommended reading that may be helpful in answering the questions:

Back

Proceed

Overview

Project: [My Research](#)
Catalog: [Software Management Plan](#)
[Reload page](#)
[Back to my projects](#)

Progress

1 of 32

Navigation

Grey entries will be conditionally skipped based on your input.

General (1 of 11)
→ Topic (1 of 5)

[Software Project Partner\(s\)](#)
[Software Project Schedule](#)
[Software Project Management](#)
[Software Development Requirements](#)

[Technical](#)
[Quality Assurance](#)
[Release and Publish](#)
[Legal and Ethics](#)

RDMO NFDI4ING – Live Demo

Abschluss Teil I

Gibt es Fragen?

RDMO NFDI4ING Support:

rdmo@nfdi4ing.de

Teil II

Best Practice im Umgang mit Forschungsdaten

Fragen? Erfahrungen? Anregungen?

Wie gut konnten Sie mit RDMO einen DMP für Ihr Projekt erstellen? (z.B. einfach, schwierig, unklar)

Was hat Ihnen bei der Nutzung besonders geholfen – oder gefehlt?

Würden Sie RDMO für Ihr Projekt weiter nutzen? Warum (nicht)?

Abschluss

Sind noch Fragen offen geblieben?

Für den nächsten Termin

- ☑ **Motivation und Grundlagen**
16.03.2026, 17:00
- ☑ **DMPs und Best Practices**
30.03.2026, 17:00
- ☐ **Metadaten und Semantik**
13.04.2026: 17:00
- ☐ **Software**
27.04.2026, 17:00

NFDI4ING RDM Basics for Engineers. Metadaten und Semantik

13. April 2026

17:00 – 18:30 Uhr

<https://tu-darmstadt.zoom-join.me/j/64207833395?pwd=Nsf8ZrCZ4gBuAOLPpKAA3Lnag5lFL0.1>

Meeting-ID: 642 0783 3395

Kenncode: 257752

Abschluss/Organisatorisches

Feedback

Mündlich

Gerne hier direkt im Kurs

Schriftlich und anonym



Sprache: DE

<https://umfragen.tu-darmstadt.de/448885?lang=de-easy>



Sprache: EN

<https://umfragen.tu-darmstadt.de/448885?lang=en>

Kleine Hausaufgabe

Bitte bis zum nächsten Kurs bei Coscine anmelden: <https://about.coscine.de/>

Voraussetzung: Zugang über Institutslogin oder alternativ via ORCID

Im nächsten Vortrag wird darin gearbeitet werden. Sollte es zu Problemen bei der Anmeldung kommen, bitte meldet euch bei uns: geschaeftsstelle@nfdi4ing.de

Thank you.

Dr. Isabel Hohle

Informationstechnologie, Forschung
und Entwicklung

ULB Darmstadt

isabel.hohle@tu-darmstadt.de

Michaela Leštáková M.Sc.

Institut für Fluidsystemtechnik

TU Darmstadt

michaela.lestakova@tu-darmstadt.de

Acknowledgement

We would like to thank the
Federal Government and the Heads of Government
of the Länder, as well as the Joint Science
Conference (GWK), for their funding and support
within the framework of the NFDI4Ing consortium.

Funded by the German Research Foundation (DFG) -
project number 442146713.

Constructing tomorrow with data



NFDI4ING

Research data management
in engineering sciences

NFDI4ING's RDM Basics for Engineers

Teil 2 von 4: **Best Practices in Research Data Management: DMPs, RDMO und Best Practice im Umgang mit Forschungsdaten**

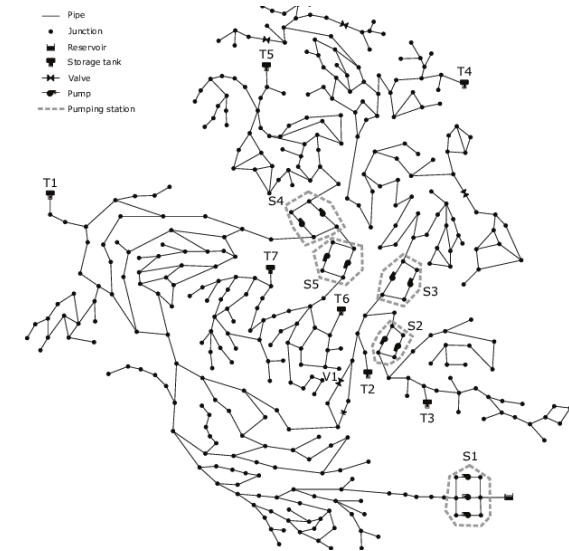
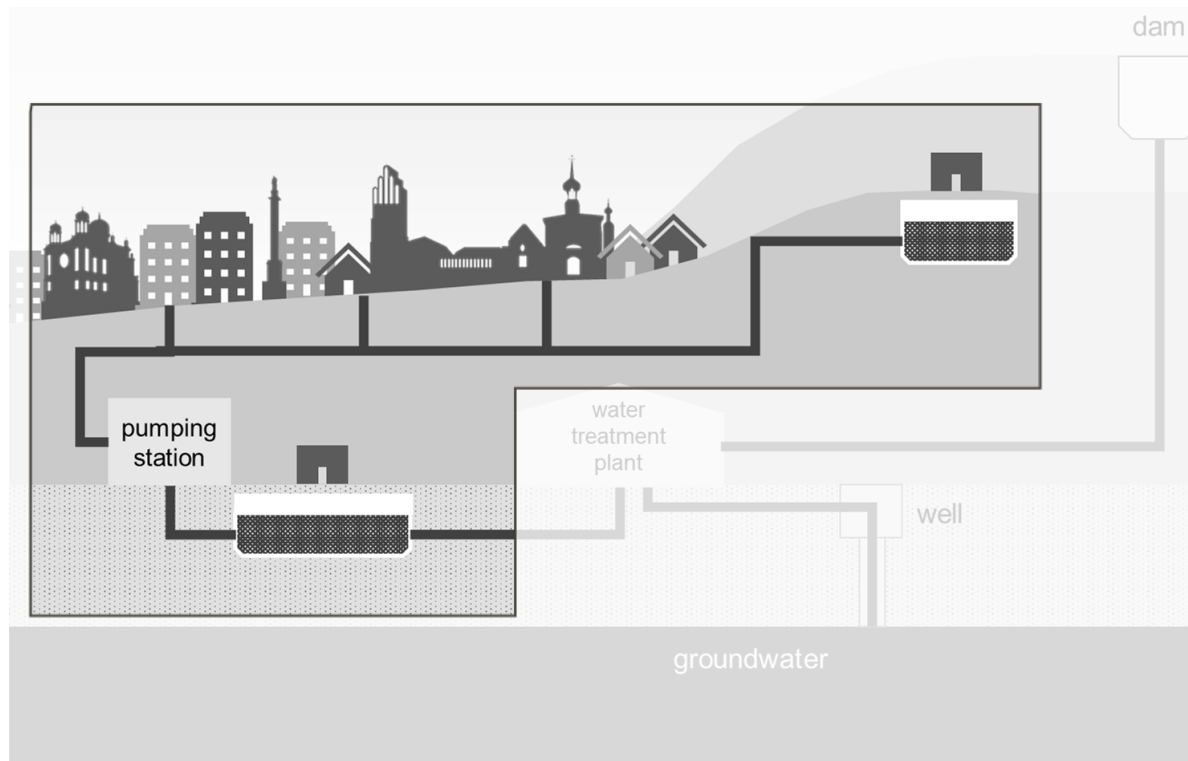
30.03.2026, online

NFDI4ING (National Research Data Infrastructure for Engineering Sciences)

Isabel Hohle, Michaela Leštáková
ULB Darmstadt/ TU Darmstadt

Trinkwasserversorgungssysteme

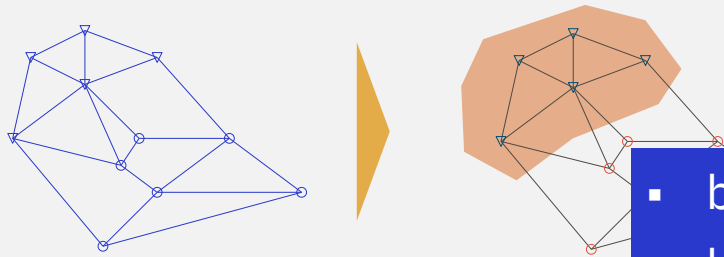
Modellierung



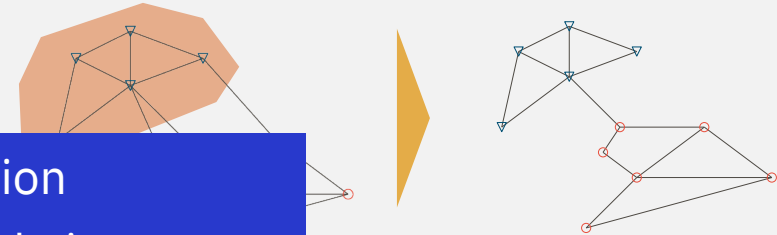
E, Pournaras & Taormina, Riccardo & M, Thapa & Galelli, Stefano & Palleti, Venkata & Kooij, Robert. (2019). Cascading Failures in Interconnected Power-to-Water Networks.

Sektorisierung der Wasserversorgungssysteme

I) NODE CLUSTERING STAGE



II) DIVIDING STAGE

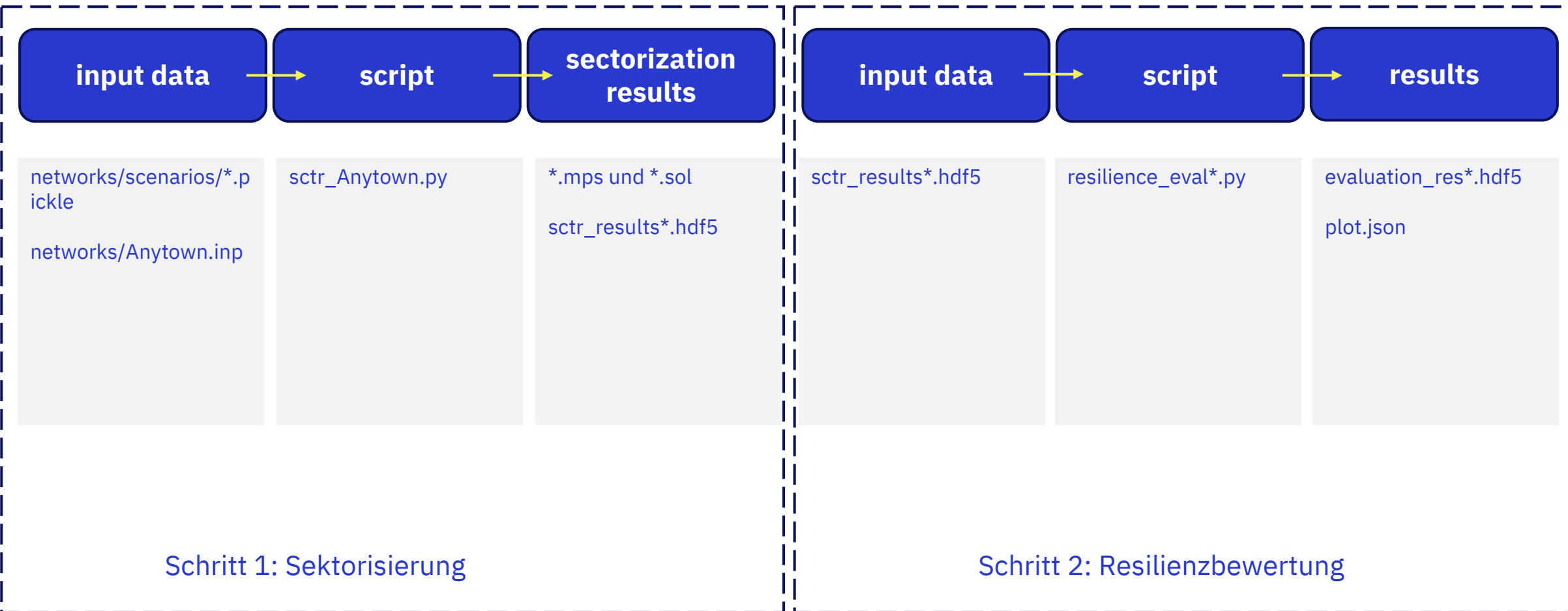


- bessere Fehlerdetektion
- bessere Überwachbarkeit
- besseres Druckmanagement
- **Resilienz?**

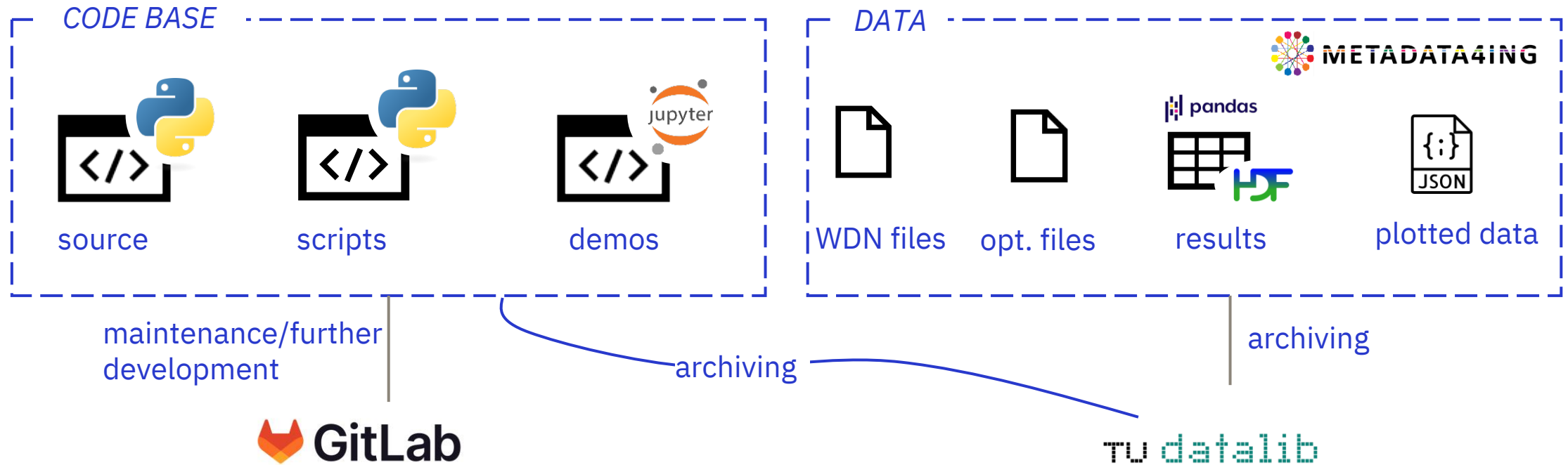
imization program

Methoden



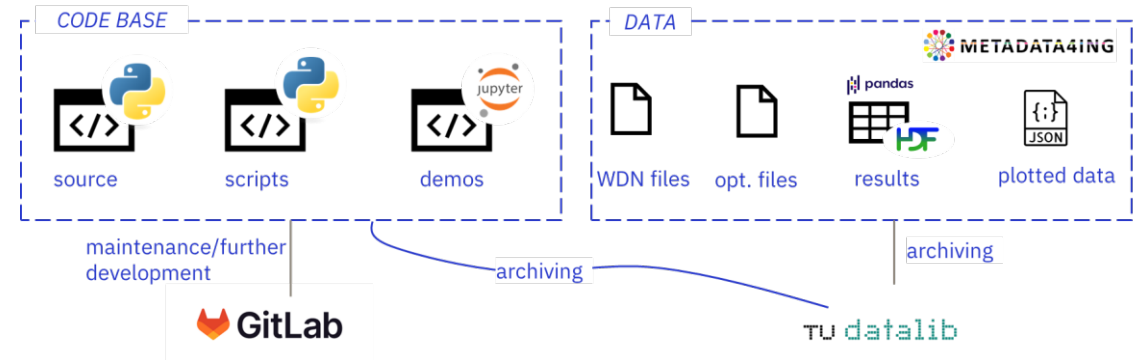


Konzept für Forschungsdatenmanagement



FDM: wichtigste Bausteine

1. Versionskontrolle für Code
2. Daten- und Codedokumentation mittels Metadaten 
3. Daten- und Codedokumentation mittels Text/Publicationen 
4. Offene Lizenzierung der Daten & Code
5. Speichern & Archivierung von Daten und Code in geeigneten Repositorien 



Resilience Assessment of Water Distribution Networks

Description	No description available.	
Catalog	NFDI4ING DMP Template (Version 2) Catalog according to the DFG checklist supplemented by engineering content.	

Tasks

Tasks are generated automatically from the answers given in the project. On the page of each task you can see which of your answers lead to the activation of the task.

No active tasks found.

Members

Here you can see who can access the project and invite additional members. You can use the user roles to manage which rights the users have. Unless you are the last owner, you can leave the project with the button next to your name.

User	E-Mail	Role	
Michaela Lestáková	michaela.lestakova@tu-darmstadt.de	Owner	

Snapshots

Snapshots allow you to save all responses at a given point in time and preserve a certain stage of the project. Later the snapshot can be used to create views, and the project can also be reset to a previous snapshot if needed.

Snapshot	Description	Created	
Test snapshot		March 27, 2026, 1:54 p.m.	  

Options

Answer questions

View answers

[Update project information](#)

[Update project catalog](#)

[Update parent project](#)

[Copy project](#)

[Delete project](#)

[Add member](#)

[Create snapshot](#)

[Back to projects overview](#)

Export

[RDMO XML](#)

[CSV \(comma separated\)](#)

[CSV \(semicolon separated\)](#)

[JSON](#)

[as ROCrate JSON](#)

Import values

Import from file



Contact

rdmo@nfdi4ing.de

Based on software RDMO funded by DFG.



This service is provided by University and State Library Darmstadt

[Imprint / Privacy Policy / Nutzungsbedingungen](#)



NFDI4ING is supported by the DFG under project number 442146713.



Update project information

Catalog

The catalog which will be used for this project.

☒ **NFDI4ING DMP Template (Version 2)**

Catalog according to the DFG checklist supplemented by engineering content.

☐ **Software Management Plan**

This catalogue is for the management of scientific software projects. It supports scientists in the development and project organisation of software developments through fifty questions in different topic blocks. This is version 3.0 of the catalogue.

☐ **Project specific: SPP2419**

This catalog was created for the DFG SPP2419 HyCAM.

Save project

Cancel

[My Projects](#) / [Resilience Assessment of Water Distribution Networks](#) / [Introduction](#)

Please add a dataset to your plan

The present catalog is the [checklist published by the DFG](#) supplemented by engineering content. The questions from the DFG checklist must be taken into account when submitting an application and completing the project and should help you to structure your data management.

During the interview, you will be asked several questions about the various aspects of data management. Depending on your research project, you may work with different types of data, so it might be necessary to plan different measures for each. Therefore, create as many sets as it makes sense for your approach.

A logical unit for your research data could be, for example, based on

- the methodology (e.g. measurements, eye-tracking, luminance measurement, questionnaires),
- data type (e.g. images, measured data, geological data), or
- accessibility of the data (public data, data of industry partner).

You must create at least one set.

Please fill in the form for each tab. The same tabs may be used later on other pages. You can add a new tab using the green button. Once created, you can edit or delete tabs using the buttons in the top right corner.

network files (.inp)

optimization files (.mps, .sol)

result files (.hdf5)

scenario files (.pickle)

rocrate (.json)

[+ Set](#)



Back

Proceed

Overview

Project: [Resilience Assessment of Water Distribution Networks](#)

Catalog: NFDI4ING DMP Template (Version 2)

[Reload page](#)

[Back to my projects](#)

Progress

76 of 82

Back

Proceed

Navigation

Grey entries will be conditionally skipped based on your input.

Introduction

→ Please add a dataset to your plan

Data description ✓

Documentation and data quality ✓

Storage and technical backup during th...

Data exchange and permanent accessib...

Legal obligations and framework condit...

Responsibilities and resources ✓

network files (.inp)

optimization files (.mps, .sol)

result files (.hdf5)

scenario files (.pickle)

[+ Set](#)

Which data types, in terms of data formats, are created in your project?

When choosing the file format, it is advisable to use formats that are standardized, non-proprietary and generally used in engineering. Please keep in mind that the data can be used for more than 10 years and should not lead to conflicts in collaborative use or subsequent use.

When it comes to the question of origin, the initial creation of data is meant, as well as the data during the project period.

Please enter your entries line by line. You can add lines using the green button and remove them using the blue cross (x).

(Comma-separated values) *.csv

(JSON) *.json

(Hierarchical Data Format) *.hdf5

(Python) *.py

[+ Entry](#)

How is the data processed in your project?

The EPANET .inp files contain descriptions of benchmark water distribution networks. They represent different operating points for each network.

To what extent are these incurred for your data set or what data volume can be expected for the data set?

Please also state the unit of measure.

☒ For each collection/measurement/scan or similar: 10KB☐ Overall, I expect a data volume of around:☐ not yet defined

network files (.inp)

optimization files (.mps, .sol)

result files (.hdf5)

scenario files (.pickle)

rocrate (.json)

[+ Set](#)

Which data types, in terms of data formats, are created in your project?

When choosing the file format, it is advisable to use formats that are standardized, non-proprietary and generally used in engineering. Please keep in mind that the data can be used for more than 10 years and should not lead to conflicts in collaborative use or subsequent use.



NFDI4ING

[Back to project](#)[Feedback](#)

data during the project period.

Please enter your entries line by line. You can add lines using the green button and remove them using the blue cross (x).

(Comma-separated values) *.csv

(JSON) *.json

(Hierarchical Data Format) *.hdf5

(Python) *.py

[+ Entry](#)

How is the data processed in your project?

The .hdf files contain results from each stage: sectorization and assessment.

The results are stored as multiindex Pandas Dataframes using pandasHDFstore.

To what extent are these incurred for your data set or what data volume can be expected for the data set?

Please also state the unit of measure.

☒ For each collection/measurement/scan or similar:

5MB

☐ Overall, I expect a data volume of around:☐ not yet defined[Back](#)[Proceed](#)

What digital methods and tools (e.g. software) are required to use the data?

In order to be able to reuse data, in addition to the data itself, the software, devices, etc. and knowledge of special procedures for use are required. Since use can vary greatly depending on the sub-area in engineering, please exchange ideas with your colleagues.

{more}As with the formats, the more standardized, open and established they are, the easier it is usually to reuse them. The establishment of standards for methods, the use of software, the collection of research data and the description of research results is an essential prerequisite for the comparability and transferability of research results.

☒ Software: Python, version:

☐ Software: Origin, version:

☐ Software: Excel, version:

☐ Software: Matlab, version:

☐ Software: Ansys, version:

☐ Self-developed software:

☒ Other software:

☐ Other digital methods:

☐ Other instruments associated with the use of certain software or methods:

Are quality controls planned and if so, how?

Quality controls can be done in different ways. They can be heavily dependent on your engineering sub-discipline, on your structures at the institute or in the project. It is important that the quality control is independent of the measures taken and refers to the product "data".

-  
- ☐ Data collection - instrument calibration (accuracy/scale)
 - ☐ Data collection - multiple measurements, observations or samples
 - ☐ Data collection - verification by experts
 - ☐ Data collection - using standardised methods and protocols (with clear instructions)
 - ☐ Data collection - computer-assisted data collection (standardised, Verified, consistency)
 - ☐ Digitization and data entry - validation rules or input masks
 - ☐ Digitization and data entry - Controlled vocabularies, code lists, and pick lists to minimize manual data entry
 - ☐ Digitization and data entry - Detailed labeling of variable and record names
 - ☐ Digitization and data entry - using a purpose-built database structure to organise data /data files
 - ☒ Digitization and data entry - Accompanying notes and documentation about the data
 - ☐ Data verification - double check of data and outlier values
 - ☐ Data verification - checking completeness
 - ☐ Data verification - Verifying random samples against the original data
 - ☐ Data verification - Check for double entry of data
 - ☐ Data verification - Statistical analyses such as frequencies, means, ranges or clustering to detect errors and anomalous values
 - ☐ Data verification - Correcting errors made during transcription
 - ☒ Data verification - Peer review
 - ☐ Other:

Data storage during the project

Please fill in the form for each tab. The same tabs may be used later on other pages. You can add a new tab using the green button. Once created, you can edit or delete tabs using the buttons in the top right corner.

network files (.inp)

optimization files (.mps, .sol)

result files (.hdf5)

scenario files (.pickle)

rocrate (.json)

+ Set



How is the data stored and backed up during the project period?

During the project, different storage locations for the actively used data may be used. Note that depending on the storage location, your data is not automatically secured.

You can find out which storage media have which advantages and disadvantages on www.forschungsdaten.info.

☒ Central solution (e.g. server in the computer center of a university or research institute):



Institut für Fluidsystemtechnik, T

☐ Decentralized solution (e.g. network disks and automatic data mirroring, own server structure at the institute or for your collaborative project, etc.):

☐ Cloud solution:

☒ GitLab: <https://git.rwth-aachen.de/disser>

☐ GitHub:

☐ Bitbucket:

☐ Own computer/laptop hard drive

☐ External hard drives

☐ Other:

network files (.inp) optimization files (.mps, .sol) result files (.hdf5) scenario files (.xml)

[+ Set](#)

Does this data set lend itself to subsequent use in other contexts? 

Please consider whether your data is of interest beyond your project and can be reused, recreated.

[... \(show more\)](#)

☐ Yes, re-use of the data by researchers in the following disciplines is conceivable:

☐ Yes, subsequent use of the data for training purposes is conceivable.

☐ Yes, my research is socially relevant and the data can be reused by society.

☐ Yes, because:

☒ No, I can't imagine a future use scenario for the data.

☐ No, because:

According to which criteria is this data record selected in order to make it available for subsequent use by others?

- ☐ Basis of scientific communication (According to the Good Research Practice, this data must be kept for 10 years)   
- ☐ For reproduction or verification of scientific results
- ☒ For reproduction or verification of scientific results
- ☐ Uniqueness (data cannot be reproduced)
- ☐ Cost / effort (Data production is difficult or expensive)
- ☒ As part of a scientific text publication
- ☒ Better traceability of the associated text publication
- ☐ High data quality
- ☐ High degree of novelty of the data
- ☐ Other:

When can this data set be used by third parties?

The *DFG Guidelines for Research Data Management* Accordingly, research data should be made available as soon as possible and ideally follow the *FAIR principles*. The importance of data publication is underscored once again by the *Guideline 13 of the Code of Good Scientific Practice*. Discipline-specific practices are pointed out here.

For the publication of research data, it is recommended to use subject-specific repositories. [... \(show more\)](#)

- ☐ In all probability already during the project.  
- ☒ Immediately after the end of the project
- ☐ 1 year after the end of the project
- ☐ 2 years after the end of the project
- ☐ 5 years after the end of the project
- ☐ 10 years after the end of the project
- ☐ Subsequent years after the end of the project:
- ☐ The data record will not be usable for third parties.

Archiving

Please fill in the form for each tab. The same tabs may be used later on other pages. You can add a new tab using the green button. Once created, you can edit or delete tabs using the buttons in the top right corner.

network files (.inp)

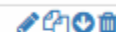
optimization files (.mps, .sol)

result files (.hdf5)

scenario files (.pickle)

rocrate (.json)

[+ Set](#)



Are you planning to archive your data in a suitable infrastructure?

In accordance with good scientific practice, research data must generally be stored for 10 years (see [Guideline 17](#) of the Code of Good Scientific Practice).

In order for the data to remain available and findable, it must also be maintained beyond the project.... [\(show more\)](#)

☒ Yes ☐ No



If yes, how and where?

☒ Archive service of your university or research institution:

TUdatalib



☐ Archive service of a partner institution:

☐ Cross-engineering archive services such as RADAR:

☐ Other:

Are there any blocking periods?



Responsibilities

Who is responsible for the adequate handling of the research data (description of the roles and responsibilities within the project)?

Possible roles are, e.g. project leader, data collector, data manager, data steward. You can find more examples on the [Harvard Medical School RDM Website](#)

Please enter your entries block by block. You can add blocks using the green button and remove them using the blue cross (×).

Who is responsible for the adequate handling of the research data (description of the roles and responsibilities within the project)? 

- ☒ First name:
- ☒ Family name:
- ☒ Affiliation:
- ☒ ORCID:
- ☒ E-Mail:
- ☒ Phone number:
- ☐ Role in the project:

+ Block

Who is responsible for curating the data after the project has ended?

Please enter your entries block by block. You can add blocks using the green button and remove them using the blue cross (×).

Who is responsible for curating the data after the project has ended? 

- ☒ First name:
- ☒ Family name:
- ☒ Affiliation:
- ☐ ORCID:
- ☒ E-Mail:
- ☐ Phone number:
- ☒ Role in the project:

+ Block

Resources

What resources are required to implement adequate handling of research data in the project? 

This can involve technical or IT resources as well as expertise that is brought in, for example, by data management or IT experts.

In case of additional costs for the use of infrastructure resources, please also enter these in the text field.

- ☐ infrastructure resource: archives:
- ☒ infrastructure resource: institutional repository:
- ☐ infrastructure resource: data base:
- ☐ infrastructure resource: RDMO:
- ☐ infrastructure resource: file server:
- ☐ infrastructure resource: cloud solution:
- ☐ infrastructure resource: licenses:
- ☐ infrastructure resource: virtual server:
- ☒ infrastructure resource: GitLab:
- ☐ infrastructure resource: Sharepoint:
- ☒ The following additional infrastructure resources are required:
- ☐ The usual infrastructure resources of the workplace are sufficient.
- ☐ Human resources through advice and/or training from the RDM team at your University or research institute:
- ☐ Personnel expenses by the institute:
- ☒ Personnel expenses in the project:
- ☐ Other:

Answers for *Resilience Assessment of Water Distribution Networks*

In the following, we have summarized the information about the project as given by you and your collaborators.

Introduction

Please add a dataset to your plan

Data description

Content data description

What kind of dataset is it?

Set "network files (.inp)": textual data

Set "optimization files (.mps, .sol)": textual data

Set "result files (.hdf5)": textual data

Set "scenario files (.pickle)": textual data

Set "rocrate (.json)": textual data

How is new data created in your project?

Snapshots

Current

→ Test snapshot

Options

Back to project overview

Export

PDF

Rich Text Format

Open Office

Microsoft Office

HTML

Markdown

mediawiki

LaTeX

Wie ist es mit Software?

Catalog

The catalog which will be used for this project.

☐ **NFDI4ING DMP Template (Version 2)**
Catalog according to the DFG checklist supplemented by engineering content.

☒ **Software Management Plan**
This catalogue is for the management of scientific software projects. It supports scientists in the development and project organisation of software developments through fifty questions in different topic blocks. This is version 3.0 of the catalogue.

☐ **Project specific: SPP2419**
This catalog was created for the DFG SPP2419 HyCAM.

Topic

If you are looking for an introduction to the topic before completing a software management plan, then we recommend Martinez-Ortiz et al. (2022): Practical guide to Software Management Plans, v1.0, <https://doi.org/10.5281/zenodo.7589725> .

What is the title of the software project?

The title of the software can, of course, still change or be specified in the course of a project. Nevertheless, it makes sense to give the project a specific name at the beginning in order to facilitate further (internal) communication about it.

Resilience Assessment of Water Distribution Networks 

Which research field(s) does this software belong to?

The list of disciplines follows the [subject classification of the DFG \(German Research Foundation\)](#).

Please enter your entries line by line. You can add lines using the green button and remove them using the blue cross (x).

Engineering Sciences / Systems Engineering    | 

 Entry

In which application class is the software categorised?

The application classes are essentially based on the "DLR Software Engineering Guidelines" (<https://doi.org/10.5281/zenodo.1344612>), p. 7-8. Depending on the selection in this questions a selection of different questions is provided below.

► [Click here for more information on the application classes for research software](#)

- ☒ Application class 0: The focus of the software is on personal use in conjunction with a small scope. The distribution of the software within and outside the own institution is not planned.  
- ☐ Application Class 1: The software is only develop within a narrow scope. It is to be further developed and used beyond personal purposes.
- ☐ Application Class 2: The software is intended to ensure long-term development and maintainability. It is the basis for a transition to product status.
- ☐ Application Class 3: For the software it is essential to avoid errors and to reduce risks. This applies in particular to critical software and that with product characteristics.

Third Party Components and Libraries

Does the software refer to other software projects or objects? 

Research software sometimes contains references to external data objects and other software solutions that are required for its execution. This aspect is particularly in request for the [FAIR4RS](#) principles I2 and R2.

Gurobi, Metis and Python packages listed in the requirements.txt file



Back

Proceed

Metadata

How do you assign metadata for your software?

For the publication of software, it is strongly recommended to include metadata (see [FAIR4RS F1 to F2 and A2](#)). [Guideline 7 in the DFG Code "Guidelines for Safeguarding Good Scientific Practice"](#) also stipulates that "[t]he source code of publicly available software must be persistent, citable and documented". If no publication is planned, it can still be useful to enrich the software with metadata. This can be particularly helpful for long-term storage at the institute, to make it easier to use the software at a later stage.

For example with a [.cff file](#) you can provide metadata in a human- and machine-readable way for your software. With the [Software Description Ontology](#) you can also use an ontology to formally describe the software. And this [handout on the relevance of software citations](#) once again collects the essential points on this complex of topics.

.cff file



Back

Proceed

License

Under what kind of license(s) will the software be published?

There are good arguments for assigning a license. [Guideline 13 in the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice"](#) even explicitly states that if "self-developed research software is to be made available to third parties, an appropriate licence is provided". [FAIR4RS R1.1](#) also emphasises the same. Please also consider the compatibility with already integrated third-party libraries.

The following websites give you an initial overview of software and licences:

- <https://choosealicense.com>
- <https://github.com/readme/guides/open-source-licensing>
- <https://opensource.org/licenses>
- <https://tldrlegal.com>

☐ AGPL-3.0-only

☐ Apache-2.0

☐ BSD-3-Clause

☐ CDDL-1.0

☐ GPL-3.0-only

☐ LGPL-3.0-only

☒ MIT

☐ Other Software License:

Back

Proceed

Fazit

- ✓ DMP gibt mir eine Übersicht der wichtigsten Punkte, die ich beim Forschungsdatenmanagement von Anfang an beachten soll
- ✓ DMP hat mir gezeigt, welche FDM-Aspekte meinem Fördergeber wichtig sind
- ✓ DMP vereinfacht den Knowhow-Transfer und Offboarding

Übung

Probiert es aus!

Erstellt einen Daten- und/oder Softwaremanagementplan für ein geplantes oder laufendes Projekt.

rdmo.nfdi4ing.de

Bei Fragen stehen wir gerne zur Verfügung.

Sind noch Fragen offen geblieben?

Vielen Dank.

Michaela Leštáková

Wissenschaftliche Mitarbeiterin

NFDI4ING Task Area Alex

TU Darmstadt

Mail: michaela.lestakova@tu-darmstadt.de

Danksagung

Wir möchten uns bei Bund, Ländern und bei der Gemeinsamen Wissenschaftskonferenz (GWK) für die Förderung und Unterstützung im Rahmen des Konsortiums NFDI4Ing bedanken.

Gefördert durch die Deutsche Forschungsgemeinschaft (DFG) - Projektnummer 442146713.

This work has been co-funded by the LOEWE initiative (Hesse, Germany) within the emergenCITY center [LOEWE/1/12/519/03/05.001(0016)/72.

The authors gratefully acknowledge the computing time provided to them on the high-performance computer Lichtenberg II at TU Darmstadt, funded by the German Federal Ministry of Education and Research (BMBF) and the State of Hesse.